Deep Learning Triad

Theory, implementation, baselines, and measured evaluation

Deep Learning | Dr. Anup Kumar Maurya | Academic Year 2026-27

TEAM: [NAME + ROLL] x 6 - human input required before submission

Assignment objectives and contribution

Questions

- How does attention support sentiment classification?
- How does a variational latent space trade reconstruction for regularization?
- How much QA error comes from retrieval versus answer extraction?

Contribution

- Three modalities: text classification, image reconstruction, grounded QA.
- Disjoint splits, fixed seed, baselines, runtime provenance, and saved predictions.
- Fast smoke runs plus full experiments on one reproducible GPU environment.

Reproducibility and compute evidence

Item	Verified configuration
Python	3.11.15
PyTorch / CUDA	2.11.0+cu128 / 12.8
GPU	NVIDIA GeForce GTX 1660 Ti (6 GiB)
Seed	42 for all three experiments
Validation	4 unit tests + compileall + result schema checks

Integrity gates

- No test examples used for fitting or tuning.
- No answer text enters the QA retrieval index.
- Fast and full outputs archived separately.
- Metrics and plots written only after successful execution.

One workflow, three different evaluation lenses

CLASSIFICATION

IMDb text -> DistilBERT -> accuracy / F1

GENERATION

Fashion image -> ConvVAE -> BCE / KL / MSE

TOOL WORKFLOW

Question -> retrieve -> extract -> EM / token F1

Scores are interpreted within each task; raw values are not compared across incompatible metrics.

Transformer foundation: scaled dot-product attention

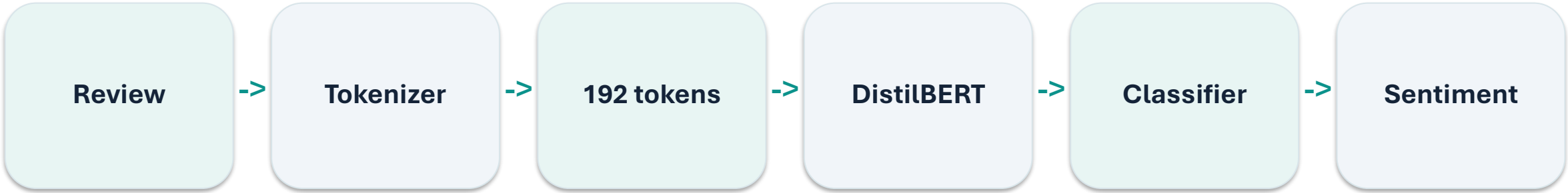
Attention(Q, K, V) = softmax(QK^T / sqrt(d_k)) V

- Q asks what each token seeks; K identifies what tokens contain; V carries content.
- Multi-head attention learns several token relationships in parallel.
- DistilBERT is a compact pretrained encoder fine-tuned with a binary head.

Why pretrained?

- IMDb fast mode has only 800 training reviews.
- Transfer learning supplies broad language representations.
- Fine-tuning adapts those representations to positive/negative sentiment.

Transformer experiment architecture



AdamW | lr 2e-5 | dynamic padding | 2 full epochs | seed 42

IMDb protocol: disjoint train, validation, and test

Mode	Train	Validation	Test	Epochs
Fast	800	200	200	1
Full	20,000	5,000	5,000	2

Dataset: stanfordnlp/imdb (25k original train + 25k original test)

Leakage control

- Train/validation: non-overlapping ranges after seeded shuffle of original train.
- Test: sampled only from IMDb original test split.
- Model and threshold choices never use test labels.

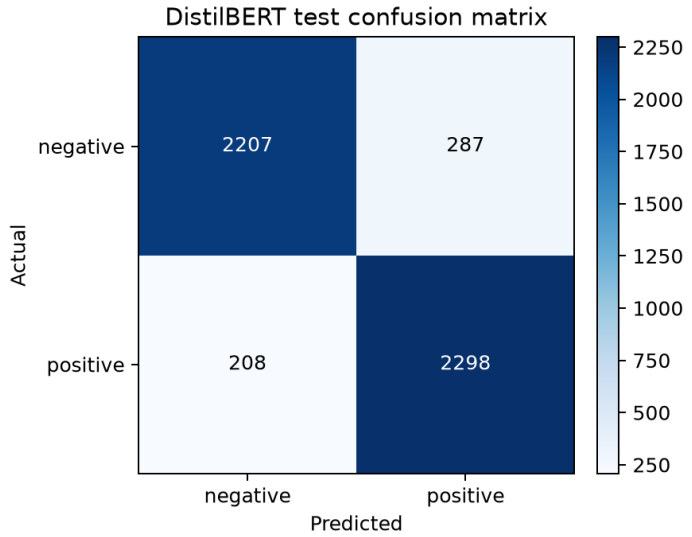
Transformer implementation: actual training path

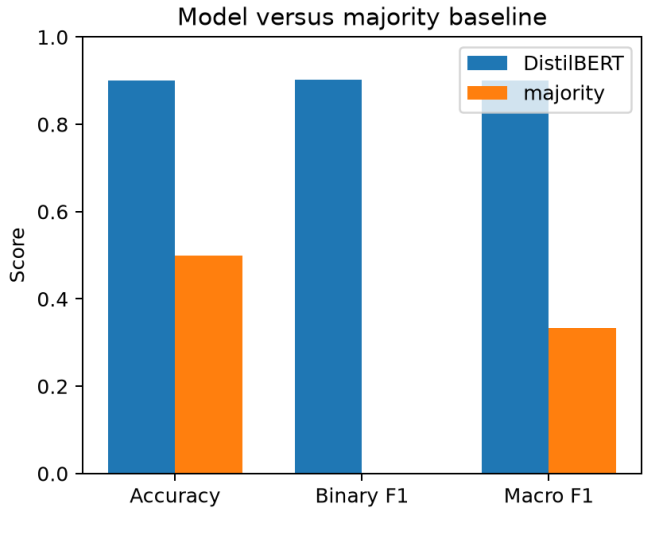
```
args = TrainingArguments(
    num_train_epochs=2, learning_rate=2e-5,
    seed=42, data_seed=42, full_determinism=True
)
trainer = Trainer(
    model=model, processing_class=tokenizer,
    data_collator=DataCollatorWithPadding(tokenizer),
    compute_metrics=metric_fn
)
trainer.train()
test_prediction = trainer.predict(test_ds)
```

Quality additions

- Binary F1 plus macro F1.
- Majority-label baseline on the same test set.
- Confusion matrix and classification report.
- Python, library, GPU, elapsed time stored in JSON.

Transformer full results: 5,000 held-out reviews

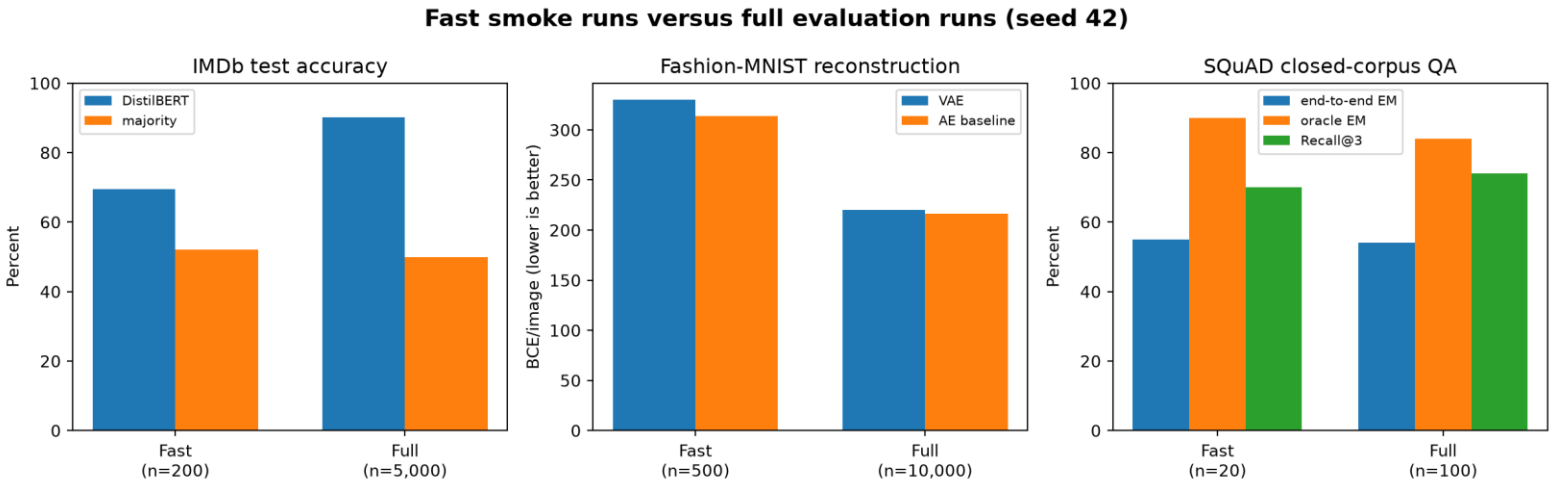




Measured results

- Accuracy: 90.10%
- Binary F1: 90.28%
- Macro F1: 90.10%
- Majority accuracy: 49.88%
- Confusion matrix: [[2207, 287], [208, 2298]]
- Train + evaluation: 1064.6s

Fast versus full runs: scale changes the evidence



Interpretation

- IMDb accuracy rose from 69.5% to 90.1%.
- Full VAE training greatly reduced reconstruction error.
- QA fast/full differences show why n=20 is only a smoke test.
- Baselines contextualize each task; metrics remain task-specific.

VAE foundation: probabilistic latent representation

Encoder and reparameterization

q_phi(z|x) -> mu, log(sigma^2)

z = mu + sigma * epsilon, epsilon ~ N(0, I)

- Sampling remains differentiable through mu and sigma.
- The decoder reconstructs x from the latent z.

Objective (negative ELBO)

Loss = BCE(x, x_hat) + beta * KL(q || N(0,I))

- BCE rewards pixel reconstruction.
- KL regularizes the latent distribution.
- beta = 1 in both fast and full runs.

Convolutional VAE architecture



Deterministic baseline: same convolutional encoder/decoder shape, one 16-D bottleneck, no KL term.

Fashion-MNIST protocol and split integrity

Mode	Train	Validation	Test	Epochs
Fast	2,000	400	500	2
Full	50,000	10,000	10,000	12

Both VAE and AE baseline see exactly the same examples and epoch count.

Protocol

- Seeded permutation of original 60k train.
- Adjacent non-overlapping train/validation subsets.
- Original 10k test remains untouched.
- Pixels scaled to [0,1]; batch 128; Adam lr 1e-3.

VAE implementation: actual loss and baseline

```
                h = self.enc(x)
        mu, logvar = self.mu(h), self.logvar(h)
z = mu + torch.randn_like(mu) * torch.exp(0.5 * logvar)
            recon = self.dec(self.fc(z))

      rec = BCE(recon, x, reduction='sum') / batch
kl = -0.5 * sum(1 + logvar - mu**2 - exp(logvar)) / batch
               loss = rec + beta * kl
                  loss.backward()
```

Fair comparison

- VAE evaluation uses latent mean mu for deterministic reconstruction.
- AE baseline optimizes reconstruction BCE only.
- Compare reconstruction BCE/MSE, not AE loss with VAE ELBO.
- Full weights saved outside Git under outputs/.

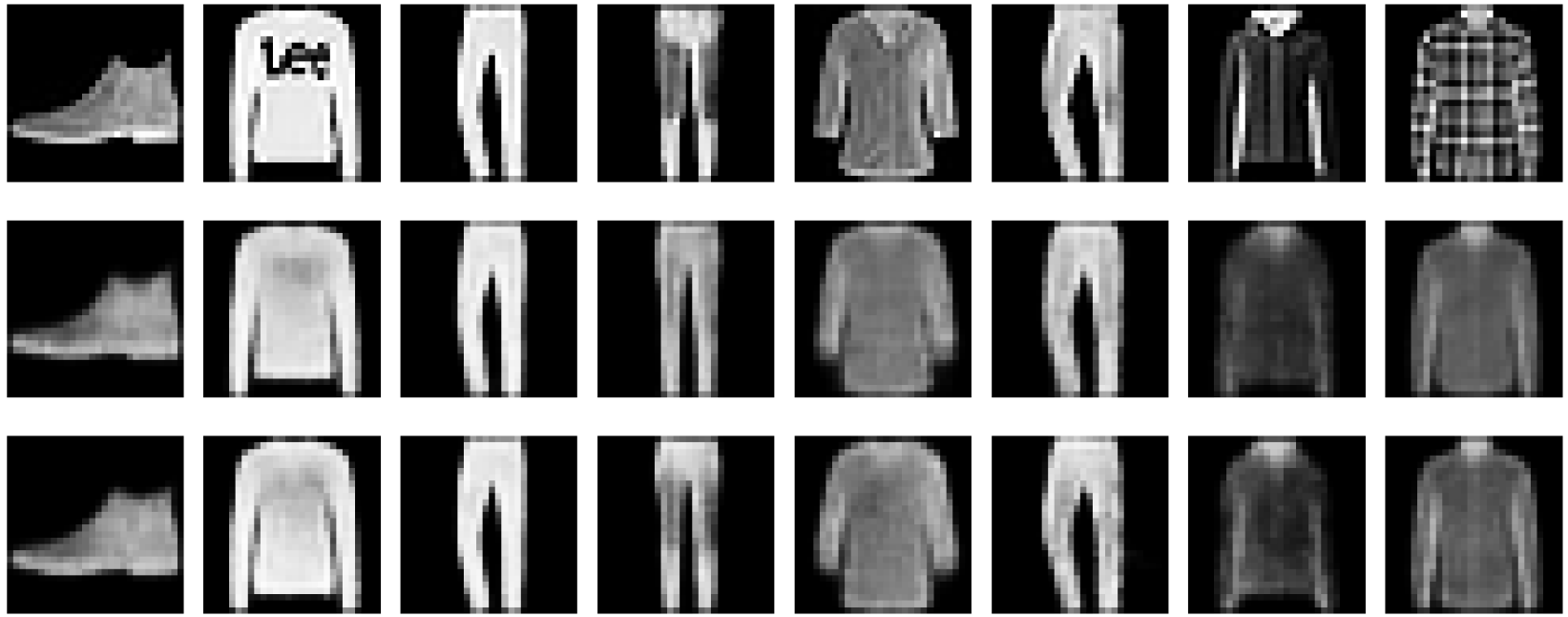
VAE full results and deterministic baseline

Metric (test n=10,000)	ConvVAE	ConvAE baseline
Reconstruction BCE / image	220.06	216.22
MSE / pixel	0.01261	0.01093
KL / image	17.47	not applicable
ELBO loss / image	237.53	not comparable

Finding

- AE wins pure reconstruction, as expected.
- VAE pays a KL cost to organize a sampleable latent distribution.
- Both models completed 12 epochs; total script time 396.3s.

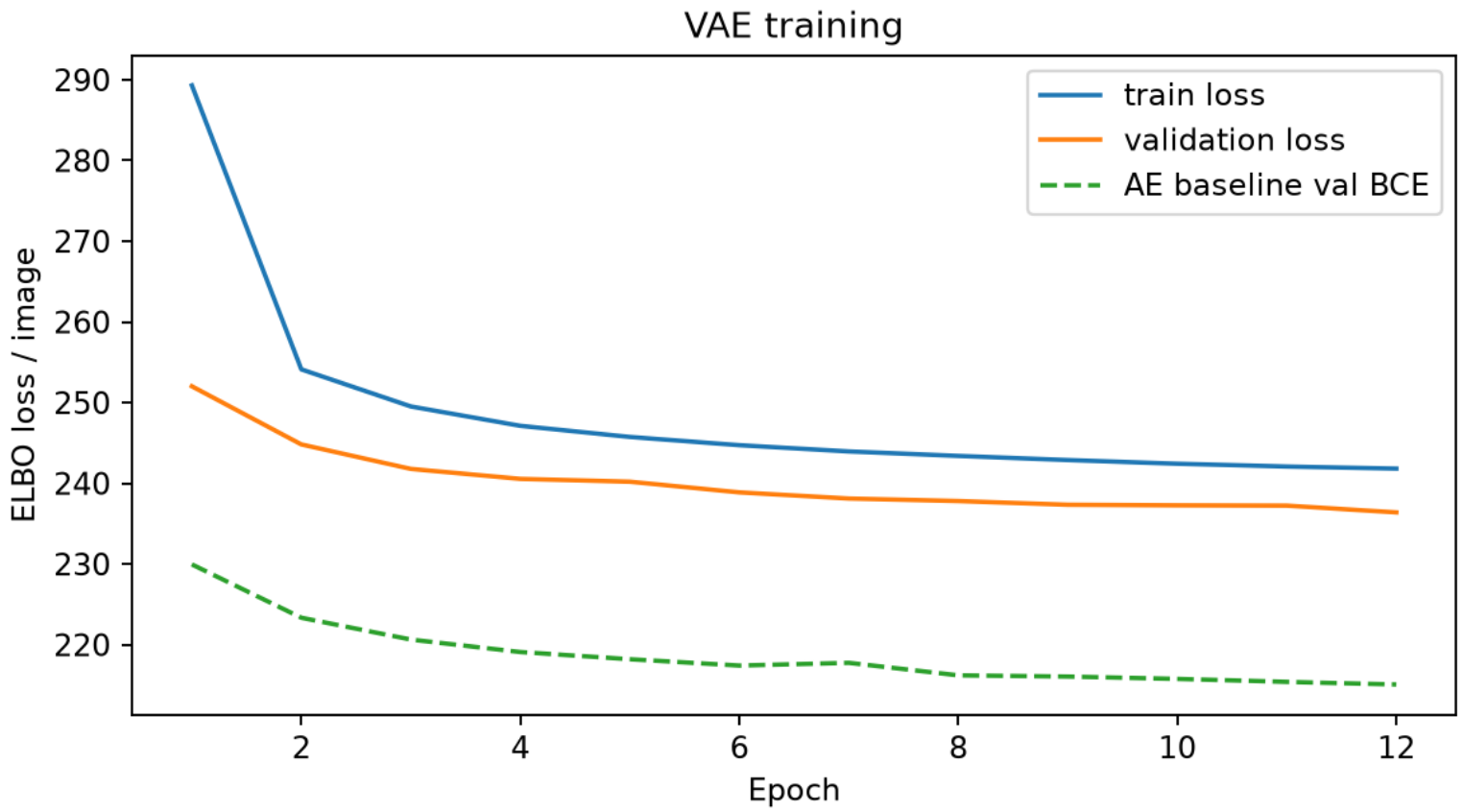
VAE qualitative evaluation: originals and reconstructions



What to inspect

- Garment silhouette is retained.
- Fine texture and sharp edges are smoothed.
- AE baseline is slightly sharper, consistent with lower BCE/MSE.
- Visuals complement, not replace, test metrics.

VAE learning curves: convergence and diminishing returns



Reading the curves

- Largest improvement occurs in early epochs.
- Validation VAE ELBO falls from 252.02 to 236.39.
- Train and validation trends remain close: no strong overfitting signal.
- AE curve is reconstruction BCE, not ELBO.

Agentic AI: a transparent tool-use workflow

Implemented agent

- Deterministic router recognizes bounded arithmetic.
- TF-IDF retrieves top-3 passages from a closed corpus.
- Extractive QA proposes spans; score combines relevance and QA confidence.
- Returns source ID, scores, excerpt, or abstention.

Claim boundary

- Not an autonomous LLM planner.
- Not web search or open-domain research.
- A lightweight, auditable orchestration pipeline.
- Calculator uses restricted AST operations - never eval().

Agentic QA pipeline and provenance



candidate score = QA confidence x max(TF-IDF relevance, epsilon)

Threshold 0.05 controls abstention before extraction.

SQuAD closed-corpus evaluation and leakage control

Mode	Questions	Unique contexts	Evidence corpus
Fast	20	2067	all validation contexts
Full	100	2067	all validation contexts

Oracle baseline isolates extraction quality from retrieval quality.

Fairness

- Documents are visible because retrieval requires a corpus.
- Answer strings are never indexed or used for routing.
- Gold context is supplied only to the explicitly labelled oracle baseline.
- EM and normalized token F1 score accepted answers.

Agent implementation: retrieval, extraction, and evidence

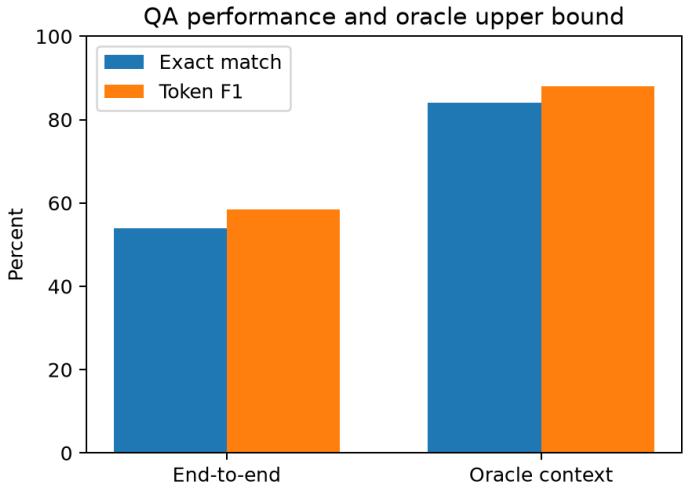
```
        hits = self.retrieve(question, k=3)
                    for h in hits:
  out = self.qa(question=question, context=h['context'])
    score = float(out['score']) * max(h['score'], 1e-9)
            candidates.append((score, out, h))

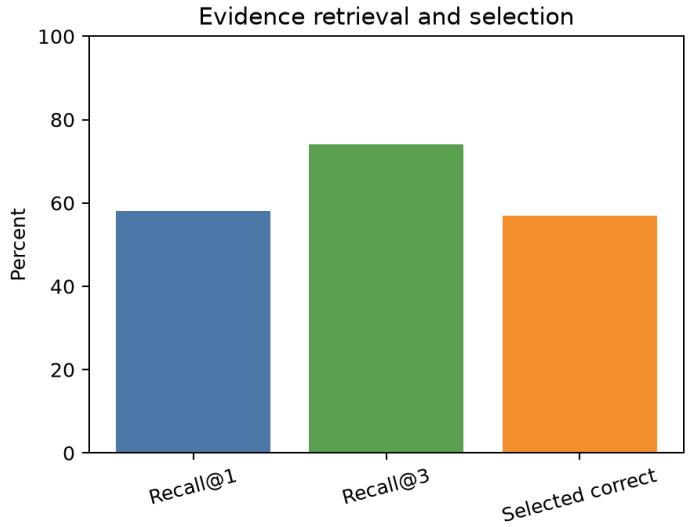
_, answer, winner = max(candidates, key=lambda x: x[0])
                    return {
                'answer': answer['answer'],
                'source_id': winner['id'],
              'retrieval_score': winner['score'],
              'qa_score': float(answer['score'])
                    }
```

Saved per question

- Prediction and accepted answers.
- Exact match and token F1.
- Top-1/top-3 gold-context indicators.
- Selected-context correctness.
- Retrieval score, QA score, source excerpt, oracle prediction.

Agent full results: retrieval is the main bottleneck





100-question evaluation

- End-to-end EM: 54.0%
- End-to-end token F1: 58.54%
- Gold-context Recall@3: 74.0%
- Selected-context accuracy: 57.0%
- Oracle-context EM: 84.0%
- Oracle-context token F1: 87.93%
- Abstention: 0.0%

Cross-experiment conclusions, limits, and next steps

What worked

- Pretrained attention clearly beat the majority baseline.
- VAE learned useful reconstructions and a regularized latent space.
- Explicit evidence tracking exposed retrieval failure separately from QA failure.

Limitations

- Single seed and one GPU environment.
- IMDb truncation can miss long-range evidence.
- Pixel BCE encourages smooth reconstructions.
- TF-IDF lexical matching and uncalibrated abstention.

Next steps

- Multi-seed confidence intervals.
- Transformer length/learning-rate ablations.
- beta and latent-dimension sweeps.
- Dense retrieval and threshold calibration.

References, recording link, and final human inputs

Core references

- Vaswani et al. (2017), Attention Is All You Need.
- Sanh et al. (2019), DistilBERT.
- Kingma & Welling (2014), Auto-Encoding Variational Bayes.
- Maas et al. (2011), Learning Word Vectors for Sentiment Analysis.
- Xiao et al. (2017), Fashion-MNIST.
- Rajpurkar et al. (2016), SQuAD.

Complete before submission

- Replace six team name/roll placeholders.
- Insert verified video URL.
- Grant viewing access to required faculty emails.
- Attach PPT/PDF and code; CC all members.
- Submit by 15 November 2026.

VIDEO LINK: [INSERT VERIFIED GOOGLE DRIVE / ONEDRIVE URL]